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Intelligent recognition method for drilling conditions of deep formation based on LSTM neural network

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Abstract

Deep formation drilling is characterized by high investment and risk. Real-time identification of drilling conditions is fundamental and prerequisite for improving drilling efficiency and reducing complex incidents. Traditionally, drilling operations relied primarily on manual post-analysis, which suffers from subjectivity and arbitrariness. This approach fails to promptly and accurately reflect actual field conditions, making it difficult to ensure timeliness and accuracy. Aiming to achieve real-time and accurate automatic identification of drilling conditions and enhance drilling efficiency, this study innovatively introduced machine learning into the entire process of identifying working conditions in deep formation drilling. By comprehensively analyzing the long-term time-series characteristics of drilling data, the authors established a real-time intelligent identification model based on a Long Short-Term Memory (LSTM) neural network. Eight drilling parameters, such as bit depth, well depth, hook height, weight on bit, weight on hook, torque, rotary speed, and standpipe pressure, were selected as input values. A 16-hidden-layerx80-node LSTM neural network model was developed. The model achieved a remarkable accuracy rate of up to 93.62% on the training set and 93.05% on the testing set. It fully demonstrated its high efficiency and reliability. LSTM attained the highest values across these metrics, with scores of 0.93, 0.91, and 0.88, compared with CNN, RNN, and SVM. Application results indicate that the model performs optimally in identifying drilling conditions. It enables real-time intelligent recognition of drilling conditions, improves drilling efficiency, and aligns with the requirements for digital and intelligent development in oilfields. This provides important theoretical and technical support for the efficient identification of drilling working conditions.

Keywords Drilling, Identification of drilling condition, Machine learning, Long short-term memory (LSTM) neural network, Intelligent prediction

1 Introduction

The shift in oil and gas exploration toward deeper formations and more complex strata has resulted in increasingly intricate downhole environments. This complexity has consequently increased the number of drilling incidents, presenting serious threats to



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drilling safety [1, 2]. However, current methods for analyzing drilling conditions typically rely on manual post-event assessments, which are inherently subjective and suffer from operational latency. The term “drilling working conditions” denotes the specific operational procedures and corresponding states of drilling equipment during field operations, including typical activities such as making a connection, tripping, back reaming, and circulating [3, 4]. The high-risk, high-investment context of deep formation drilling imposes stringent demands for operational efficiency and low incident rates. Consequently, the accurate, real-time identification of these conditions forms the essential foundation for improving drilling efficiency and mitigating complex incidents. In actual deep drilling operations, condition analysis is predominantly conducted through manual, post-event assessment, which introduces subjectivity and arbitrariness. This method cannot promptly or accurately reflect actual downhole conditions, thereby compromising both the timeliness and accuracy of the evaluation. Traditional approaches that rely on experience-based formulas and logical reasoning often lead to extended non-productive time and reduced drilling efficiency. Although recent studies have developed physics-based or empirical models for automated condition identification, the highly complex and nonlinear relationships among monitoring parameters continue to pose significant challenges to achieving both rapid and precise recognition. Consequently, critical abnormal events are often addressed with delayed responses. In contrast, machine learning methods are capable of effectively modeling the high-dimensional, nonlinear mappings between various monitoring parameters. As a data-driven “black-box” approach, they perform pattern recognition tasks without explicit programming, thereby offering a powerful and adaptive solution to these persistent challenges.

Yu HY identified prediction points for drilling conditions by analyzing trends in single-parameter drilling data and developed multi-condition recognition criteria using an expert system with trend regression lines, achieving effective judgment of drilling conditions [5]. Guan ZC applied fuzzy mathematics to establish a fuzzy discrimination model for drilling condition recognition [6]. Wang JP constructed an artificial neural network-based prediction and diagnosis model for drilling accidents to reduce their occurrence and associated costs [7]. Alsalama AM introduced an integrated system incorporating essential data mining and analysis tools, which significantly shortened early warning times compared to conventional methods [8]. Balaji K provided a comprehensive overview of the development of data-driven methods in oil and gas engineering from fundamental, theoretical, and applied perspectives [9]. Sun T proposed a method for drilling condition identification based on support vector machines (SVM), developing multiple intelligent discrimination models and deriving their optimal parameter values [10]. Arnaout A designed a mathematical model using discrete polynomials to recognize specific patterns in drilling conditions [11]. Gerald employed an extreme gradient-boosting (XGBoost) methodology to predict ECD values [12]. Fatemeh proposes an innovative machine learning-driven framework for ROP prediction, employing advanced algorithms such as Least Squares Support Vector Machines (LSSVM), Artificial Neural Networks (ANN), and Random Forest (RF) [13]. Ben et al. noted that due to top drive vibration, distinguishing between “rotary drilling” and “sliding drilling” conditions based solely on surface rotary speed is challenging, while machine learning models significantly outperform rule-based computational approaches [14]. Liu et al. identified invisible non-productive time during drilling by establishing automated drilling condition

recognition methods and calculating key performance indicators for drilling efficiency [15]. Zhang et al. developed a drilling loss risk warning model based on BP neural networks and Long Short-Term Memory (LSTM) networks. By using causal inference algorithms to interpret model warnings and incorporating risk mechanisms to implement alarm constraints, they improved the timeliness and accuracy of loss warnings [16]. We G employs multiheaded self-attention and residual connections to capture long-range temporal dependencies in drilling data, and adds a feed-forward classification head to the encoder output to recognize seven operation states [17]. The particle swarm optimization (PSO) algorithm was used to optimize the parameters of the kernel function of the support vector machine (SVM), ensuring the best learning and prediction performance [18].

Although previous studies have attempted to develop machine learning models for identifying drilling conditions, most have overlooked the long-term sequential nature of comprehensive mud logging data, leading to high false alarm rates. To address this limitation, the authors developed a real-time intelligent drilling condition recognition model based on a Long Short-Term Memory Recurrent Neural Network (LSTM-RNN). The model utilizes eight input parameters: bit depth, well depth, hook height, weight on bit, hook load, torque, rotary speed, and standpipe pressure, with drilling conditions as the output. It successfully enables intelligent identification of eight distinct drilling conditions, including rotary drilling, slide drilling, downward circulation, reaming hole, upward circulation, back reaming, tripping out, and tripping in.

2 Basic principles

An Artificial Neural Network (ANN) is a computational learning system inspired by the structure and function of biological neural networks in the human brain. It consists of numerous interconnected simple processing units that collectively form a complex network capable of learning from data. The key to achieving intelligent recognition of drilling conditions lies in selecting the most suitable neural network architecture based on the characteristics of the sample dataset. The classification performance of the machine learning model is then evaluated using standard metrics such as accuracy, precision, recall, and F1-score. The overall workflow for applying an Artificial Neural Network to predict drilling conditions is illustrated in Fig. 1.

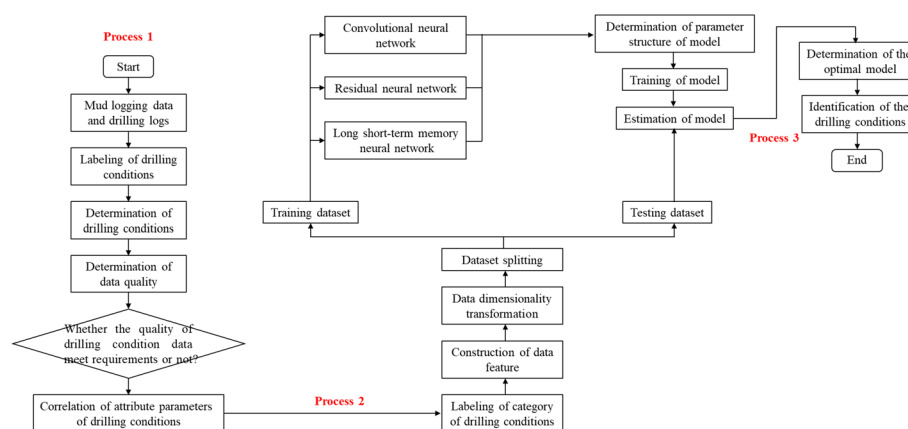


Fig. 1 The workflow of drilling condition recognition based on ANN

2.1 Long short-term memory network (LSTM)

By analyzing the architectures of BP, RNN, and LSTM neural networks, and considering the long-time-series characteristics of comprehensive mud logging data, the LSTM network was selected as the optimal algorithm for intelligent drilling condition discrimination. It excels at learning temporal patterns and processing long-range dependencies.

Conventional Recurrent Neural Networks (RNNs) can only handle certain short-term sequence problems and are unable to address long-term dependency issues. This is because when sequence data is long, gradients from the back end of sequences have difficulty propagating backwards to the front-end during backpropagation, easily leading to vanishing gradient and exploding gradient problems. Long short-term memory network, also known as LSTM recurrent neural networks, are a variant of RNNs that can effectively solve the vanishing and exploding gradient problems. They were first proposed by S. Hochreiter and J. Schmidhuber in 1997. Compared to conventional RNNs, the recurrent units in LSTM networks are more complex, incorporating a gating mechanism. This allows the weights within the recurrent units to be dynamically adjusted. Such gated recurrent units can selectively preserve useful historical information and discard irrelevant historical information. Thus, they can avoid the problems of vanishing or exploding gradients.

The basic structure of an LSTM unit is illustrated in Fig. 2. Its core cell comprises three gates: the input gate, the forget gate, and the output gate. These three gates establish a recurrent connection within the internal state of the LSTM mechanism. The reason LSTM can address the long-term dependency problem inherent in RNNs lies in its introduction of the forget gate f_t , input gate i_t , and output gate o_t . These gates regulate the flow and loss of information (features) throughout the network. The chain-like structure of an unfolded LSTM is illustrated in Fig. 2.

In Fig. 2, the long horizontal line traversing from left to right at the top allows learned knowledge to propagate directly along it without significant alteration, thereby enabling the LSTM network to possess “long-term” memory and resolve the “long-term dependency” problem inherent in RNNs. The combined utilization of short-term memory and long-term memory empowers LSTM to learn both short-term patterns and long-term patterns across extended time sequences. As the long-term memory cell state passes through the neural network, the forget gate (f_t) discards (removes) portions of

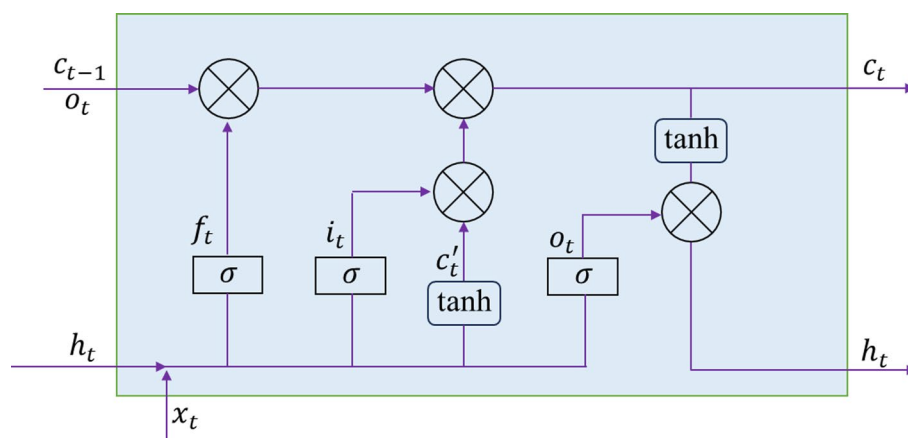


Fig. 2 Operation of the LSTM recurrent unit at time step t

memorized information, the input gate (i_t) incorporates (adds) new information, and finally, the output gate processes and filters the result.

Figure 1 illustrates the recurrent unit structure of an LSTM network at time step t . As shown in Fig. 1, the process first involves replicating the output of the hidden layer at time step $t-1$ and the input at time step t into four copies each. These copies are then assigned distinct randomly initialized weights. Next, the vectors h_{t-1} and x_t are concatenated. Finally, this concatenated vector undergoes nonlinear transformation through either the sigmoid or tanh activation function. This process yields the activation vectors for the forget gate, input gate, and output gate, as well as the transformed cell state vector. The corresponding calculation formulas are as follows:

$$f_t = \sigma(W \cdot [h_{t-1}, x_t] + b) \quad (1)$$

$$i_t = \sigma(W \cdot [h_{t-1}, x_t] + b) \quad (2)$$

$$o_t = \sigma(W \cdot [h_{t-1}, x_t] + b) \quad (3)$$

$$c'_t = \tanh(W \cdot [h_{t-1}, x_t] + b) \quad (4)$$

Where: f_t , i_t and o_t - The activation vectors for the forget gate, input gate, and output gate at time step t , respectively; c'_t - The candidate cell state vector (or new information vector) at time step t , transformed via the tanh layer; W - Weight matrix; b - Bias vector; x_t - The input at time step t ; h_{t-1} - The output of hidden layer at time step $t-1$; σ - The activation function of Sigmoid.

The forget gate f_t and input gate i_t is used to discard portions of irrelevant stored information and retain partial useful information at time step t , thereby updating the internal memory cell state c_t . The calculation formula is:

$$c_t = f_t \cdot c_{t-1} + i_t \cdot c'_t \quad (5)$$

Where: c_t - The updated memory cell state at time step t ; c_{t-1} - The preserved memory cell state from time step $t-1$.

The output gate o_t selects useful information from the cell state c_t and transmits it to the hidden state h_t . The calculation formulas are:

$$h_t = o_t \cdot \tanh(c_t) \quad (6)$$

Where: The output of hidden layer at time step t .

2.2 Framework of intelligent prediction model of drilling condition based on LSTM

The structure of the Long Short-Term Memory (LSTM) recurrent neural network resembles that of a fully connected neural network, consisting of an input layer, hidden layers, and an output layer. The hidden layers form the core of the network architecture. Theoretically, increasing both the number of neurons and the depth of the network enhances its fitting capability and prediction accuracy. The recurrent units in the LSTM model constructed in this study are inherently suited for processing sequential data and constitute a relatively deep structure by design. Accordingly, the proposed network architecture incorporates one LSTM layer as a hidden layer, followed by 20 to 30 fully connected (Dense) layers, resulting in a multi-hidden-layer topology. After finalizing the

number of layers, a K-fold cross-validation algorithm is employed during experimentation to optimize hyperparameters, such as the number of neurons in the hidden layers and the learning rate.

The intelligent drilling condition prediction model framework developed in this study consists of five main components: data preprocessing, an input layer, an LSTM layer, fully connected layers, and an output layer. During the data preprocessing phase, input variables that are strongly correlated with drilling conditions are first selected. After preprocessing, the dataset is divided into training and testing sets. The processed data are then fed sequentially into the input layer and the subsequent hidden layers, undergoing iterative computation until the model converges. The final output forms the structure of the LSTM-based recurrent neural network drilling condition prediction model, as illustrated in Fig. 3.

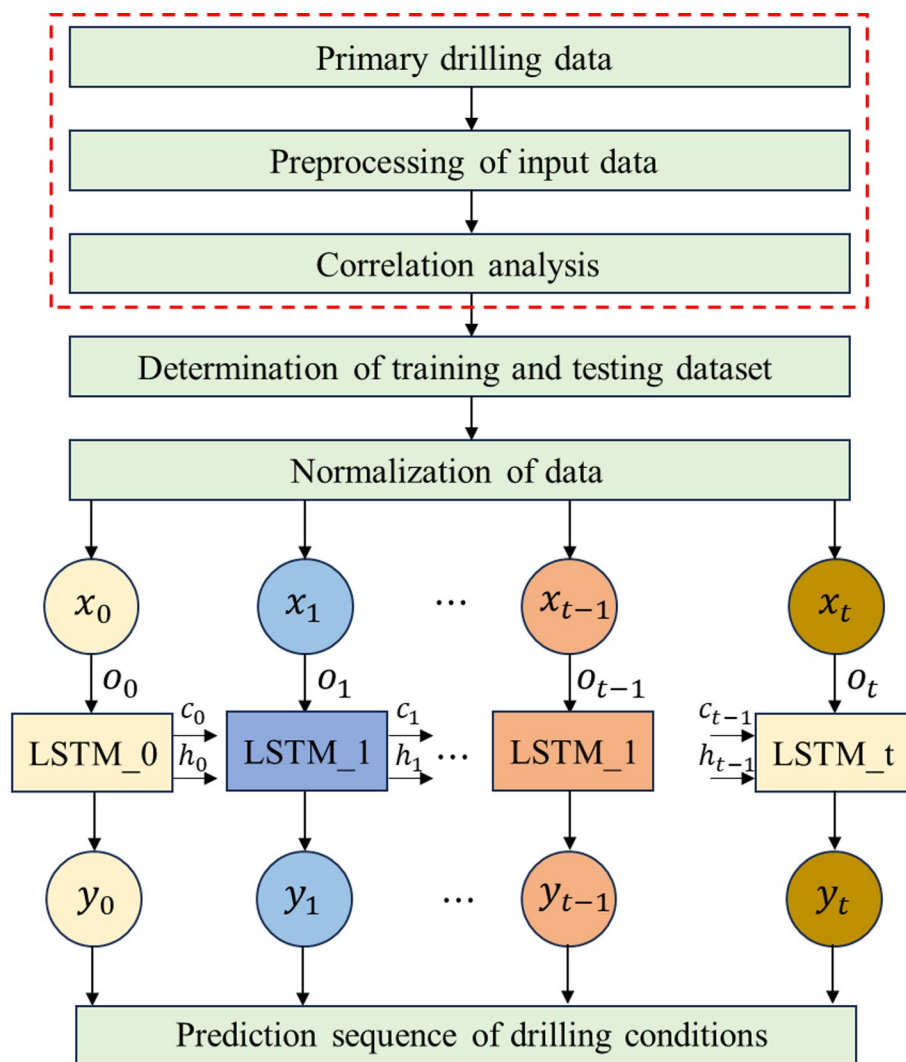


Fig. 3 The framework for intelligent prediction model of drilling condition based on LSTM Recurrent Neural Network

Table 1 Confusion matrix of binary classification results

Actual values	Predicted values	
	Positive	Negative
True	TP	FN
False	FP	TN

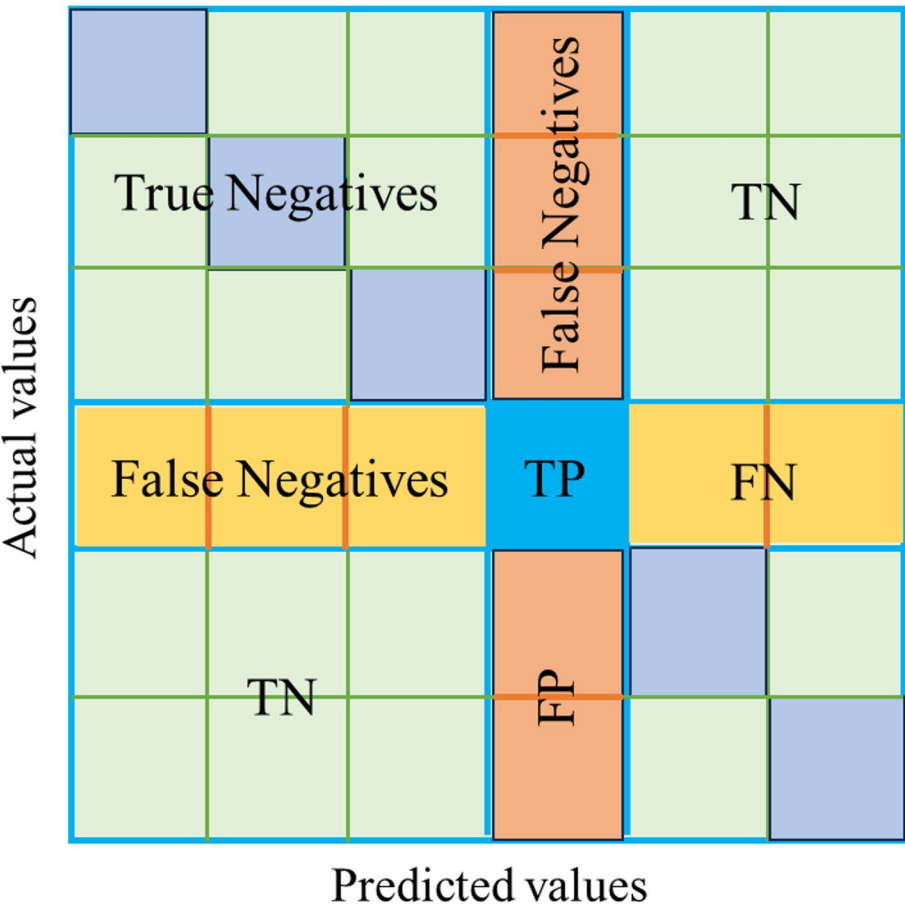


Fig. 4 Confusion matrix for multi-class classification problem

2.3 Performance evaluation metrics of machine learning model

Generally, generalization error is used to evaluate the performance of machine learning models. A smaller generalization error indicates better model performance. A test set is employed to assess the model’s classification ability, and the test error on this set serves as an approximation of the generalization error. The author utilizes four evaluation metrics, namely, accuracy, precision, recall, and F1-score.

By combining the actual class labels and the model’s predicted class labels for classification, the confusion matrix for binary classification is shown in Table 1, and the confusion matrix for multi-class classification is illustrated in Fig. 4. Here, TP stands for True Positive, FP for False Positive, TN for True Negative, and FN for False Negative.

Different metrics directly reflect the performance of classification. Accuracy is the most common evaluation criterion, i.e., the number of correctly classified samples divided by the total number of samples. For balanced classification problems, a higher accuracy usually indicates a better classifier. Precision and Recall are a pair of metrics

that reflect a trade-off. Precision indicates, for a given class, what proportion of predicted positives are correct, and what proportion are incorrect. Recall shows, among all actual positives of a class, how many were correctly predicted. The F1-score is the harmonic mean of precision and recall.

3 Sample set data and organization

3.1 The parameters of comprehensive mud logging

Comprehensive mud logging data contains numerous parameters, which can be categorized into three types, such as drilling parameters, drilling fluid parameters, and gas logging parameters. The specific classifications and contents are detailed in Table 2. Directly incorporating all comprehensive mud logging data into the model for training would lead to excessive model complexity. This not only increases the difficulty of model training but also potentially compromises the model’s output effectiveness. Drilling fluid parameters and gas logging parameters reflect the performance of the drilling fluid and the characteristics of gas, respectively. They exhibit poor correlation with drilling operating conditions. Conversely, eight drilling parameters, namely, real-time bit depth, measured depth, hook height, weight on bit, hook load, torque, rotary speed, and standpipe pressure, are closely related to drilling operating condition characteristics. For example, during compound drilling, well depth equals bit position and gradually increases, but hook height gradually decreases. Weight on bit, rotary speed, torque, flow rate, and standpipe pressure are all non-zero. This demonstrates that under different drilling operating conditions, these eight drilling parameters exhibit distinct and specific patterns. Therefore, this study selects the following eight drilling parameters from the mud logging comprehensive data as characteristic parameters for model training, namely, real-time bit depth, measured depth, hook height, weight on bit, hook load, torque, rotary speed, and standpipe pressure.

3.2 Selection of input variable

The research dataset is derived from actual drilling operations at ten wells in Xinjiang province, China, and includes over 50 types of measurements, such as rotary speed, well depth, torque, and bit position. Using mud logging parameters as the input vector for the neural network model, a series of nonlinear transformations through the hidden layers produce probability distributions for different drilling conditions at the output layer. The condition with the highest probability among all predictions is identified as the current operational state. Given the large number of available drilling parameters, using all of them as independent input features would result in an overly complex and redundant neural network architecture. Therefore, based on field expertise and the relative importance of each parameter for condition identification, a reduced set of comprehensive mud logging parameters was selected to decrease input dimensionality. The eight most representative and critical parameters, such as real-time bit depth, measured depth,

Table 2 The types of parameters of composite mud logging

Composite data	Content of data
Drilling parameters	Well depth, Bit position, Hook height, Hook load, Weight on bit, Rotary speed, Torque, Flow rate, Standpipe pressure
Drilling fluid parameters	Inflow/outflow drilling fluid density, Inflow/outflow drilling fluid temperature
Gas logging parameters	Total hydrocarbons, Methane, Hydrogen sulfide

Table 3 Statistical analysis of sample dataset

Drilling parameters	Minimum	Maximum	Mean	Standard deviation
Bit depth/m	0	5286.22	2025.32	450.35
Measured depth/m	552.00	5286.45	2621.57	398.38
Hook height/m	1.53	39.58	15.76	4.78
Weight on bit/kN	0	279.85	90.55	26.89
Hook load/kN	0	4956.79	998.53	202.75
RPM/(r·min ⁻¹)	0	175.00	70.36	15.97
Torque/(kN·m)	0	55.67	16.45	4.19
Standpipe Pressure/MPa	0	50.88	13.71	3.16

Table 4 Data distribution and One-Hot encoding for 8 drilling conditions

Drilling conditions	Data volume	Distribution of data volume (%)	Label	One-hot Encoding
Rotary drilling	9 125 378	40.09	0	100 000 00
Sliding drilling	492 549	2.16	1	010 000 00
Downward washing	890 724	3.91	2	001 000 00
Reaming	2 950 845	12.97	3	000 100 00
Upward circulation	885 973	3.89	4	000 010 00
Reverse reaming	2 659 467	11.69	5	000 001 00
Tripping out	3 225 795	14.18	6	000 000 10
Tripping in	2 525 879	11.00	7	000 000 01
Total	22 756 610	100		

hook height, weight on bit, hook load, torque, rotary speed, and standpipe pressure, were employed to accurately distinguish among eight drilling conditions: rotary drilling, slide drilling, washing down, reaming, washing up, back reaming, tripping out, and tripping in. Accordingly, the LSTM network implemented in this study uses these eight mud logging parameters as the input feature vector and the eight drilling conditions as the output vector.

3.3 Labeling of sample data

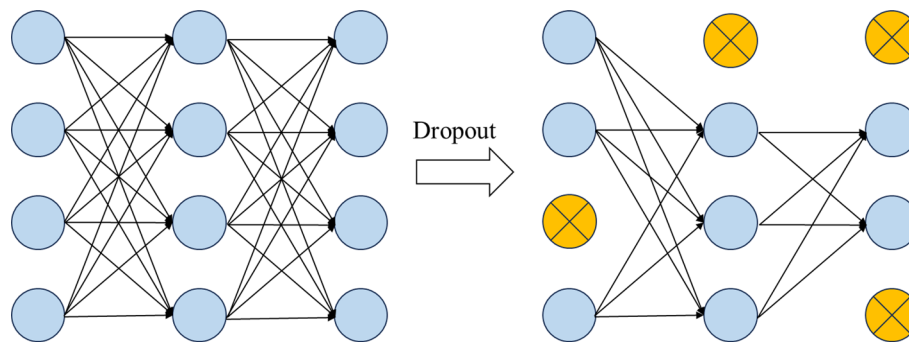
Based on the collected drilling dataset, a sample set for working condition classification was established as shown in Table 3.

The drilling process is marked by highly complex and variable downhole conditions, extended duration, subtle and often ambiguous process variations, and intricate challenges in real-time automated analysis during operations. The drilling working condition corresponding to each drilling moment serves as the output for network training. However, the drilling dataset cannot directly provide the working condition states. Therefore, before model training, manual annotation of drilling conditions must be performed by referencing drilling logs. The data distribution and labels for the eight conditions used in the classification task are shown in Table 4. It is apparent that slide drilling, downward washing, and upward circulation collectively account for less than 5% of the data distribution, highlighting a significant class imbalance in the classification task. Subsequent analysis should therefore focus on Recall metrics.

The composite logging dataset of deep formations used in this section contains a total of 20 million samples. Of these, 75% are allocated to the training set and 25% to the test set. Supervised learning algorithms are then applied to train and validate the model for the classification and prediction of drilling conditions. The manual labeling criteria for

Table 5 Labeling criteria for the eight typical drilling conditions

Input	Rotary drilling	Sliding drilling	Downward washing	Reaming	Upward circulation	Reverse reaming	Trip-ping out	Trip-ping in
Hook height	↓	↓	↓	↓	↑	↑	↑	↓
Hook load	↑	↑	Slight↓	Slight↓	Slight↑	Slight↑	↓	↑
Weight on bit	>0	>0	0	0	0	0	0	0
RPM	>0	0	0	>0	0	>0	0	0
Torque	>0	0	0	>0	0	>0	0	0
Standpipe Pressure	>0	>0	>0	>0	>0	>0	0	0
Bit depth/ Measured depth	≥	≥	<	<	<	<	<	<

**Fig. 5** Dropout mechanism of LSTM neural network

the eight typical drilling conditions are provided in Table 5. Any remaining composite logging data not corresponding to these predefined conditions is labeled as “Other”.

4 Construction of long short-term memory (LSTM) neural networks

4.1 The optimization of hyperparameter

The selection of the number of hidden layers in a neural network significantly impacts the accuracy of model training. When making this choice, it is essential to both ensure training accuracy and prevent the model from becoming “overfitted”. The number of neurons in the input layer and output layer corresponds to the number of input features (comprehensive logging parameters) and output features, respectively. While the hidden layers and the number of nodes per layer are critical, they are challenging to determine. Generally, more hidden layers and nodes can improve prediction accuracy, even approaching 100%. However, this may lead to overfitting, reducing performance on test datasets. The neural network models with 8–25 hidden layers are most commonly used in the practical applications. To address the issue of overfitting, a dropout mechanism was employed within the LSTM architecture (Fig. 5). Dropout is a standard regularization method in which, during each training iteration, a random subset of neuronal outputs is omitted. This stochastic process ensures that the network utilizes a different combination of neurons in every iteration. By preventing excessive co-adaptation of nodes on the training data, dropout significantly improves the model’s robustness and generalization ability, making it a highly valuable technique for enhancing neural network performance.

Considering the complexity of the input and output items, this study tested neural network models with 8, 16, and 25 hidden layers, each containing 10–120 neurons per

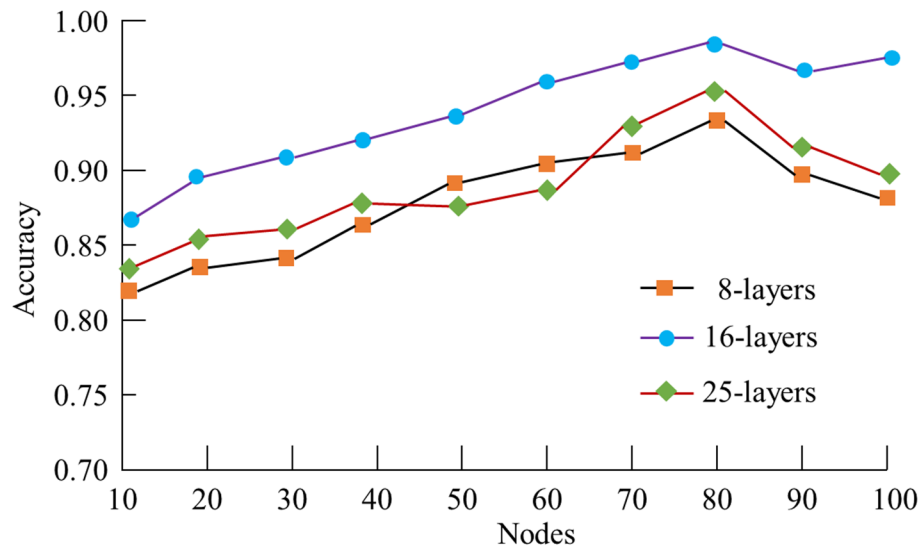


Fig. 6 The test accuracy of LSTM models with different hidden layers and node configurations

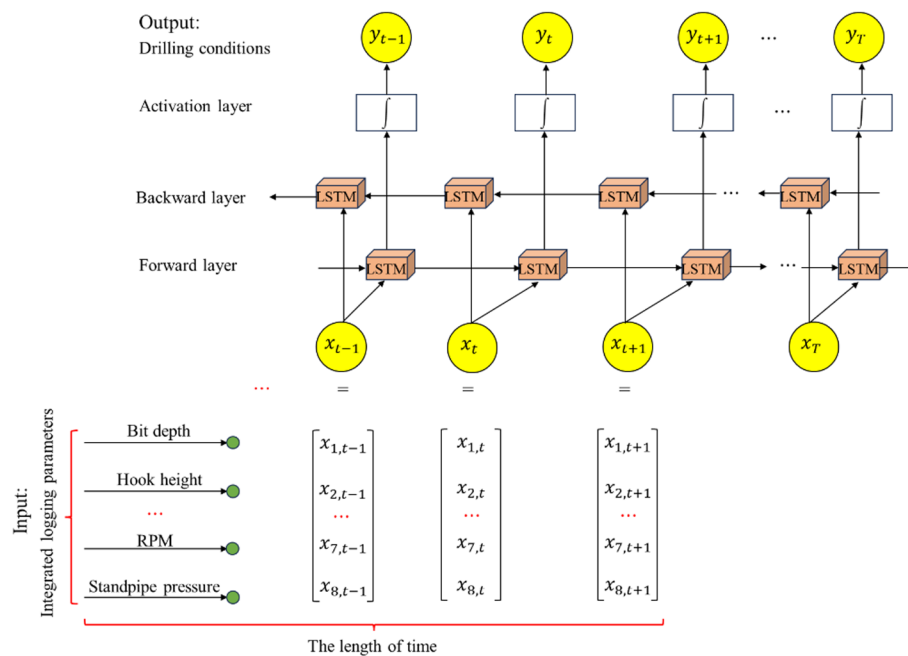


Fig. 7 Intelligent identification model for drilling conditions based on LSTM

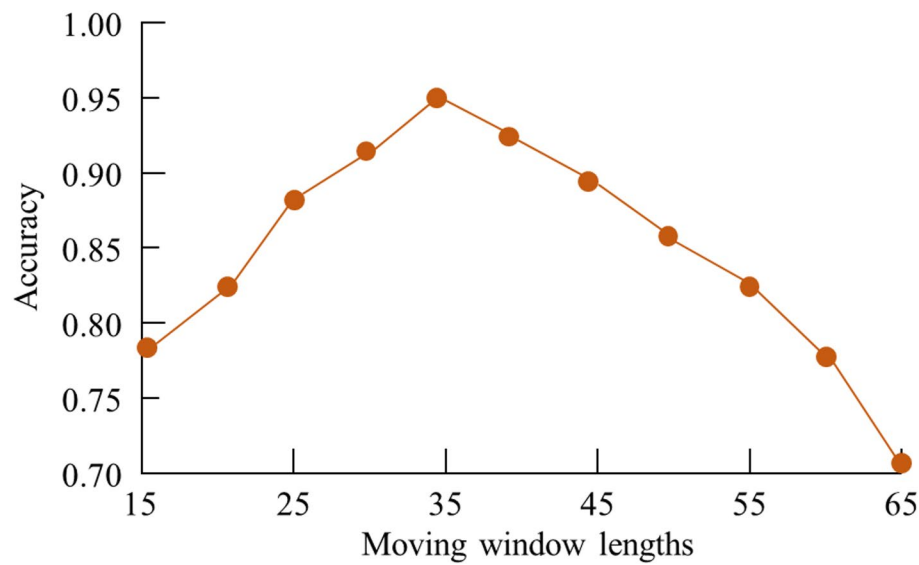
layer, totaling 25 LSTM models. As shown in Fig. 6, the model with 16-hidden layers $\times$ 80-neurons per layer achieved the highest accuracy (96.5%) on the test set. Consequently, the structure of neural network 16-layer $\times$ 80-neuron (illustrated in Fig. 7) was selected as the final model. Hyperparameter details of the LSTM model are provided in Table 6.

4.2 Moving window length

During actual field operations, the data collected by sensors exhibits stochastic fluctuations during sampling. This leads to instability and uncertainty in drilling condition classification, resulting in a phenomenon where multiple drilling operation types might be

Table 6 Hyperparameters and network architecture of the optimal LSTM model

Option	Parameters
Maximum iterations	150
Training dataset/Testing dataset	75%/25%
Optimization algorithm	Adam
The number of hidden layers	16
Number of nodes per layer	80

**Fig. 8** Classification accuracy of the LSTM network on the test set at different moving window lengths

identified within brief periods. This can potentially cause misclassification of conditions. For example, multiple drilling conditions are sometimes identified within a short span of 30 s, which is clearly unrealistic and erroneous. To mitigate or reduce data errors caused by these fluctuations, a Moving Window approach is employed in the preprocessing stage. This preprocessing method effectively smooths the data, reducing the impact of short-term variations. It more accurately reveals long-term trends and improves data analysis accuracy. Particularly in noisy environments, it offers high flexibility and adaptability, as the window size can be adjusted based on specific circumstances to reduce the instability in drilling condition data acquisition. Consequently, condition classification is determined not only by the instantaneous sensor transmission data but also by the average value calculated over the size of the moving window. As the moving window progressively shifts backward, it determines the drilling condition type for the subsequent data point. The moving window length was tested from 15 to 65 in increments of 5. As shown in Fig. 8, the results indicate that the accuracy rate increases from 15 to 35 and begins to decrease from 35 to 65. Therefore, the final data window length was selected as 35, which aligns with the requirements for field operations.

4.3 Balanced focal loss function

To improve the performance of classification models on imbalanced datasets, this paper introduces an enhanced balanced focal loss function (Bal_Focal_loss), building upon Focal Loss and Balanced Cross-Entropy. The proposed loss combines class balancing

with focal loss principles and further incorporates temperature scaling and a simulated annealing mechanism to enhance both the stability and accuracy of the model.

In imbalanced datasets, samples from minority classes are frequently underrepresented during training, resulting in a model bias toward the majority class. To address this challenge, a class-balanced weighting factor is incorporated into the loss function. Specifically, the prior probability for each class is estimated and scaled by a parameter τ to construct a class-balancing term. This term is applied to the input logits prior to the loss computation, effectively calibrating the prediction probabilities for all classes.

In classification tasks, the output of a model is conventionally represented as a probability distribution obtained *via* the **softmax** function. To calibrate the confidence of predictions, this work introduces a temperature parameter, integrated with a simulated annealing strategy, to smooth the output distribution. Furthermore, the annealing process dynamically adjusts the temperature to mitigate overconfident predictions. The corresponding formulation is provided below:

$$\text{prior} = \tau \ln \frac{1}{C} \quad (7)$$

$$T = \max(T_f, T_\eta) \quad (8)$$

$$X = \frac{x + \text{prior}}{T} \quad (9)$$

Where: prior- the class prior probability, employed to adjust the model's prediction distribution; C- The number of classes; τ - The prior adjustment parameter; T- Temperature parameter, which smooths the prediction distribution; T_f - The final temperature (at termination of annealing); η - The annealing rate, controlling the speed of temperature reduction.

In this study, the focal loss is incorporated into the loss function to enhance the model's performance on hard-to-classify samples by assigning greater weight to these samples, thereby reducing the contribution of easy-to-classify samples to the overall loss. The weight factor of the focal loss and the final loss calculation formula are as follows:

$$\text{Fcoal_weight} = \alpha(1 - p_t)^\gamma \quad (10)$$

$$\text{Loss} = \frac{1}{N} \sum_{i=1}^N \alpha_i (1 - p_{t_i})^\gamma \ln p_{t_i} \quad (11)$$

Where: α_i - The weighting factor for the i -th class; γ - The modulating factor; P_{ti} - The predicted probability of the correct class for the i -th sample; N- The total number of samples.

5 Training and performance validation of the long short-term memory neural network

5.1 Training of the long short-term memory network

Utilizing the neural network architecture shown in Fig. 6; Table 6, a 5-fold cross-validation was implemented. Each training session ran for 120 iterations, with the LSTM network trained using the Adam optimization algorithm. Evaluation metrics were subsequently calculated. As depicted in Fig. 9, the Accuracy and Loss across different

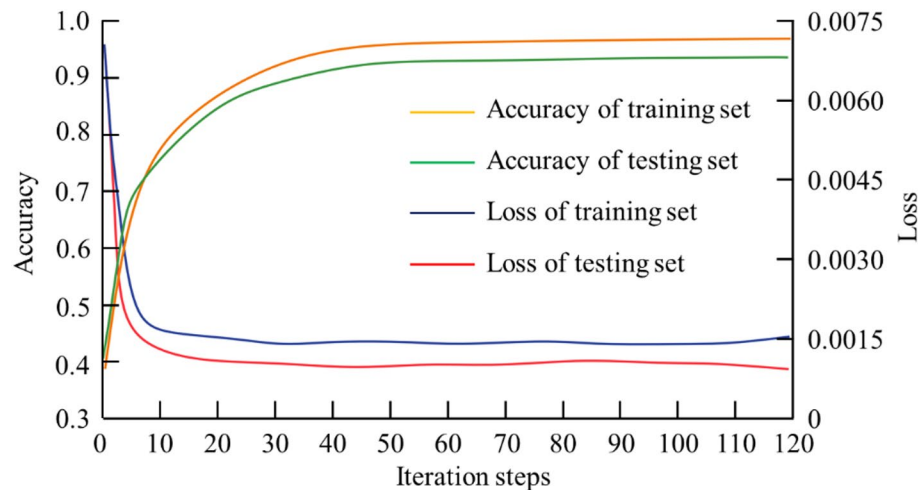


Fig. 9 Accuracy and loss during training iterations

iterations demonstrate that both the training and test sets exhibit significant improvement in accuracy throughout the 120 iterations. After 120 iterations, both the training set and test set have achieved high accuracy rates with no significant further improvement observed, indicating that the training objective has been successfully accomplished. The final accuracy reached 93.62% (training set) and 93.05% (test set), with losses of 0.00017 and 0.000202, respectively. Throughout the iterative training, the trends of accuracy and loss on the training set align consistently with those on the test set. This robust agreement confirms no significant overfitting and indicates strong generalization capability of the model.

5.2 Performance validation of the LSTM network

The overall performance metrics of the LSTM network were evaluated using confusion matrices, with results presented in Fig. 10; Tables 7 and 8. In Fig. 9 the horizontal and vertical axes represent the predicted and actual outcomes of the eight drilling conditions, respectively. As shown in Fig. 10a; Table 7, the recall rates for three critical drilling conditions-sliding drilling, stationary status, and downward well washing-reached 0.94, 0.95, and 0.96 on the training set. Figure 10b; Table 8 demonstrate that the corresponding recall rates on the test set were 0.93, 0.95, and 0.94. Although slightly lower than the training set, these values remain high (>0.93). It indicates that.

Collectively, the neural network achieves exceptional accuracy with strong generalization capability on unseen test data. Furthermore, the model completes prediction for a single sample in under 56 ms- well below the real-time data acquisition frequency of 1 Hz (1,000ms). This demonstrates true real-time identification capability, ensuring timely processing for field deployment of the LSTM model in live drilling operations. Such computational efficiency aligns with engineering constraints and fulfills practical drilling requirements.

5.3 Comparison with other models

Subsequently, the author applied the dataset to three additional machine learning models, namely, CNN, RNN, and SV, observed from Table 9. Then, compared their performance with that of LSTM. The results show that the CNN model achieved the lowest

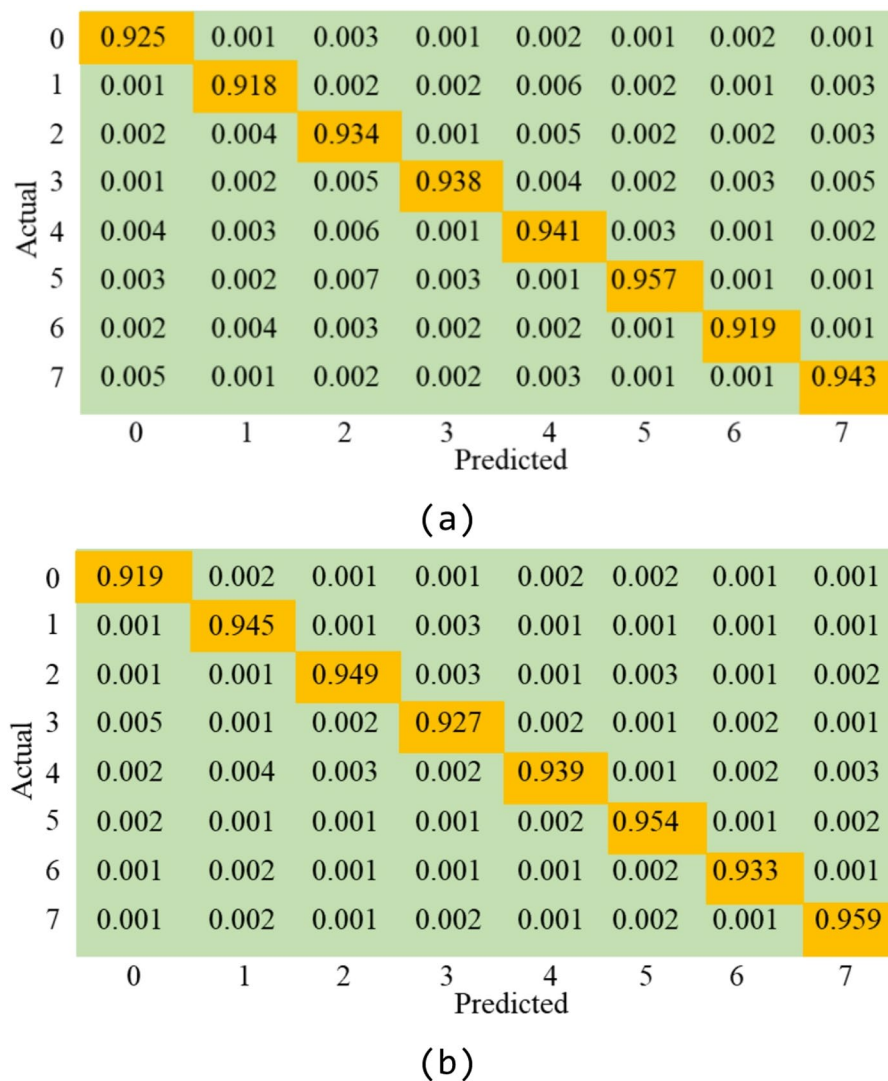


Fig. 10 Confusion Matrices of the LSTM Network. **a** Confusion Matrix on the Training Set, **b** Confusion Matrix on the test Set

Table 7 Performance metrics of the LSTM network on the training set

Drilling conditions	Precision	Recall	F1 score
Rotary drilling	0.96	0.93	0.92
Sliding drilling	0.91	0.93	0.90
Downward washing	0.84	0.92	0.91
Reaming	0.96	0.97	0.96
Upward circulation	0.91	0.96	0.92
Reverse reaming	0.93	0.97	0.93
Tripping out	0.97	0.94	0.95
Tripping in	0.94	0.92	0.95
Others	0.88	0.91	0.87

scores in recall, precision, and F1-score, at 0.78, 0.75, and 0.71, respectively. In contrast, LSTM attained the highest values across these metrics, with scores of 0.93, 0.91, and 0.88. This indicates that CNN performs the poorest among the models tested, while LSTM performs the best. These findings suggest that CNN, RNN, and SVM are limited

Table 8 Performance metrics of the LSTM network on the test set

Drilling conditions	Precision (0.01)	Recall (0.01)	F1 score (0.01)
Rotary drilling	0.91	0.94	0.95
Sliding drilling	0.93	0.91	0.96
Downward washing	0.81	0.95	0.85
Reaming	0.92	0.93	0.92
Upward circulation	0.94	0.92	0.95
Reverse reaming	0.95	0.91	0.96
Tripping out	0.93	0.92	0.97
Tripping in	0.92	0.92	0.96
Others	0.91	0.86	0.92

Table 9 Comparison between performance of different machine learning models

Model	Precision (0.01)	Recall (0.01)	F1 score (0.01)
CNN	0.78	0.75	0.71
RNN	0.81	0.83	0.79
SVM	0.86	0.82	0.81
LSTM	0.93	0.91	0.88

in their ability to handle complex temporal features, especially in tasks that require integrating information across multiple time steps, whereas LSTM demonstrates a clear advantage in such contexts.

6 Conclusions

- (1) A comprehensive real-time intelligent identification process for drilling operations in deep formations using LSTM networks was proposed. This demonstrates the unique advantages of LSTM in handling complex, high-dimensional, nonlinear mapping relationships within long time series, providing a reference for studying other classification problems during deep formation drilling processes.
- (2) Four evaluation metrics-Accuracy, Precision, Recall, and F1-score-were employed to assess the classification performance of the machine learning models. Generalization error was selected as the criterion for evaluating model performance; a smaller generalization error indicates superior model performance.
- (3) Eight comprehensive mud logging parameters-bit depth, well depth, hook height, weight on bit (WOB), hook load, rotary speed, torque, and standpipe pressure-were optimally selected as input feature vectors. An LSTM network model with 16 hidden layers and 80 nodes per layer was established. This model achieves real-time intelligent identification of 8 typical drilling operations: rotary drilling, slide drilling, making a connection, stationary, circulating, washing down, reaming, washing up, back reaming, tripping out, tripping in, and "other". It achieved a precision of 95.53% on the test set, providing a machine learning model foundation for subsequent drilling efficiency analysis and early warning of complex incidents.

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Author contributions

Le Jiao completed the conception, structure design, writing. Ping Du completed the software and data analysis. Yuan Yang completed the data processing. Xinxin Fang completed the writing and figures.

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Data availability

The datasets generated and analysed during the present study are not publicly available because of applicable institutional guidelines but are available from the corresponding author upon reasonable request.

Declarations**Ethics approval and consent to participate**

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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